

OBSTETRICS

Reductions in stillbirths and preterm birth in COVID-19—vaccinated women: a multicenter cohort study of vaccination uptake and perinatal outcomes



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BACKGROUND: COVID-19 infection in pregnancy is associated with a higher risk of progression to severe disease, but vaccine uptake by pregnant women is hindered by persistent safety concerns. COVID-19 vaccination in pregnancy has been shown to reduce stillbirth, but its relationship with preterm birth is uncertain.

OBJECTIVE: This study aimed to measure the rate of COVID-19 vaccine uptake among women giving birth in Melbourne, Australia, and to compare perinatal outcomes by vaccination status.

STUDY DESIGN: This was a retrospective multicenter cohort study conducted after the June 2021 government recommendations for messenger RNA COVID-19 vaccination during pregnancy. Routinely collected data from all 12 public maternity hospitals in Melbourne were extracted on births at ≥ 20 weeks' gestation from July 1, 2021 to March 31, 2022. Maternal sociodemographic characteristics were analyzed from the total birth cohort. Perinatal outcomes were compared between vaccinated and unvaccinated women for whom weeks 20 to 43 of gestation fell entirely within the 9-month data collection period. The primary outcomes were the rates of stillbirth and preterm birth (spontaneous and iatrogenic) in singleton pregnancies of at least 24 weeks' gestation, after exclusion of congenital anomalies. Secondary perinatal outcomes included the rate of congenital anomalies among infants born at ≥ 20 weeks' gestation and birthweight $\leq$ third centile and newborn intensive care unit admissions among infants born without congenital anomalies at ≥ 24 weeks' gestation. We calculated the adjusted odds ratio of perinatal outcomes among vaccinated vs unvaccinated women using inverse propensity score-weighting regression adjustment with multiple covariates; $P < .05$ was considered statistically significant.

RESULTS: Births from 32,536 women were analyzed: 17,365 (53.4%) were vaccinated and 15,171 (47.6%) were unvaccinated. Vaccinated

women were more likely to be older, nulliparous, nonsmoking, not requiring an interpreter, of higher socioeconomic status, and vaccinated against pertussis and influenza. Vaccination status also varied by region of birth. Vaccinated women had a significantly lower rate of stillbirth compared with unvaccinated women (0.2% vs 0.8%; adjusted odds ratio, 0.18; 95% confidence interval, 0.09–0.37; $P < .001$). Vaccination was associated with a significant reduction in total preterm births at < 37 weeks (5.1% vs 9.2%; adjusted odds ratio, 0.60; 95% confidence interval, 0.51–0.71; $P < .001$), spontaneous preterm birth (2.4% vs 4.0%; adjusted odds ratio, 0.73; 95% confidence interval, 0.56–0.96; $P = .02$), and iatrogenic preterm birth (2.7% vs 5.2%; adjusted odds ratio, 0.52; 95% confidence interval, 0.41–0.65; $P < .001$). Infants born to vaccinated mothers also had lower rates of admission to the neonatal intensive care unit. There was no significant increase in the rate of congenital anomalies or birthweight $\leq$ 3rd centile in vaccinated women. Vaccinated women were significantly less likely to have an infant with a major congenital anomaly compared with the unvaccinated group (2.4% vs 3.0%; adjusted odds ratio, 0.72; 95% confidence interval, 0.56–0.94; $P = .02$). This finding remained significant even when the analysis was restricted to women vaccinated before 20 weeks' gestation.

CONCLUSION: COVID-19 vaccination during pregnancy was associated with a reduction in stillbirth and preterm birth, and not associated with any adverse impact on fetal growth or development. Vaccine coverage was substantially influenced by known social determinants of health.

Key words: cohort studies, COVID-19, pregnancy outcome, premature birth, stillbirth, vaccination

Introduction

Pregnancy is an independent risk factor for severe COVID-19 infection, with pregnant people having higher rates of

hospitalization, intensive care unit admission, and mortality compared with nonpregnant adults.¹ COVID-19 infection during pregnancy also increases the risk of serious adverse perinatal outcomes, including stillbirth, preterm birth, cesarean delivery, and preeclampsia.

An effective vaccine against SARS-CoV-2 became available globally from early 2021,² with the Australian rollout commencing in March 2021.³ By the end of 2021, vaccine uptake in the eligible Australian population was 85 per 100.⁴ It is now well-established that the

messenger RNA (mRNA) COVID-19 vaccine is safe and protective against severe COVID-19 disease in pregnancy,^{5–8} but this confidence was not present at the launch of the vaccine program because of the exclusion of pregnant women from early clinical trials.⁹ It was not until June 9, 2021 that Australian health authorities changed their advice on vaccination in pregnancy from cautiously recommending deferral to the postpartum period, to clearly recommending vaccination with an mRNA vaccine during pregnancy.¹⁰ The change in health advice was met with vaccine

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AJOG at a Glance

Why was this study conducted?

COVID-19 infection in pregnancy is associated with a higher risk of progression to severe disease, but vaccine uptake by pregnant women is hindered by persistent safety concerns. COVID-19 vaccination in pregnancy has been shown to reduce stillbirth, but its relationship with preterm birth is uncertain. Most of the published literature on COVID-19 vaccination in pregnancy has methodological limitations, including fixed-cohort bias and time-varying exposure. We conducted this multicenter study to provide robust evidence on messenger RNA (mRNA) COVID-19 vaccination and perinatal outcomes including stillbirth, preterm birth, and congenital anomalies.

Key findings

The adjusted odds of stillbirth, preterm birth, and neonatal intensive care admission were markedly reduced among infants born to COVID-19–vaccinated women compared with those born to unvaccinated women. Both spontaneous and iatrogenic preterm births were reduced among vaccinated women. COVID-19 vaccination during pregnancy was not associated with an increase in congenital anomalies or fetal growth restriction.

What does this add to what is known?

Our analysis confirmed a strong relationship between the COVID-19 mRNA vaccine and lower stillbirths and iatrogenic and spontaneous preterm births. We provided further evidence to endorse the safety of the mRNA vaccine during pregnancy and to promote the benefits of reduced stillbirth and preterm birth. Vaccine coverage was not distributed equally in our population and was influenced by known social determinants of health. In addition to its impact on reducing severe COVID-19 illness, vaccination may have been a proxy for other biological and social determinants of health in our pregnant population.

hesitancy among some pregnant Australian women, with common concerns including side effects of the vaccine on their newborns and a perception of inadequate safety information.¹¹ Vaccination was available free of charge through a range of public vaccination centers, general practitioners, and within the antenatal clinics in some maternity services.

A recent systematic review and meta-analysis of 23 studies confirmed the effectiveness of the COVID-19 vaccine in pregnancy, and showed a significant reduction in stillbirth.¹² However, this study was inconclusive about the impact of COVID-19 vaccination on preterm birth. The meta-analysis of the 6 studies that reported on preterm birth showed a nonsignificant relative reduction in preterm birth of 10%. Only 2 studies accounted for time-dependent exposures, that is, included adjustment for the fact that the vaccination occurred at

various gestations.¹³ Women vaccinated in late pregnancy are not at risk of very preterm birth, which skews the results toward fewer preterm births in the vaccinated group. Another limitation of the literature is failure to account for fixed cohort bias.¹⁴ Fixed cohort bias occurs in retrospective cohorts that are defined by date of birth using calendar start and end dates. This has the effect of introducing both left fixed truncation bias (start of data collection occurs at a fixed date, but at variable gestations for each included pregnancy) and right censoring bias (otherwise eligible participants are not included because their birth occurs after the study end date). When the exposure rate (ie, vaccination) is higher at the end of the study period, this will have the effect of overestimating the preterm birth rate in the exposed group because term vaccinated pregnancies that deliver after the end of the cohort date will not be captured.

In Melbourne, Australia, data collection on maternal COVID-19 vaccination status was mandated by the Department of Health and Aged Care for all births from July 1, 2021. These vaccination data were collected by a collaboration of 12 Melbourne public maternity hospitals under a research protocol established to monitor the effect of the pandemic on clinical quality indicators.¹⁵ Data from the Collaborative Maternity and Newborn Dashboard (CoMaND) for the COVID-19 pandemic were used here to assess the perinatal outcomes associated with vaccination in pregnancy, including preterm birth, stillbirth, and congenital anomalies.

Materials and Methods**Institutional review board approval**

This study was given ethical approval from the human research ethics committees of Austin Health (Ref. HREC/64722/Austin-2020) and Mercy Health (Ref. 2020-031).

Study population

We extracted routinely collected data on births at ≥ 20 weeks of gestation from all 12 public maternity hospitals in Melbourne from July 1, 2021 to March 31, 2022. Data on pregnancies that ended before 20 weeks are not collected in our routine maternity data collection and were not available for this study. Approximately 80% of all hospital births in Melbourne occur in these study sites. Births in exclusively private hospitals and planned home births outside of publicly funded home-birth programs were not captured. However, women planning a private hospital or home birth would typically be transferred to a public hospital if they were at risk of preterm birth < 31 weeks or required tertiary maternal–fetal medicine care.

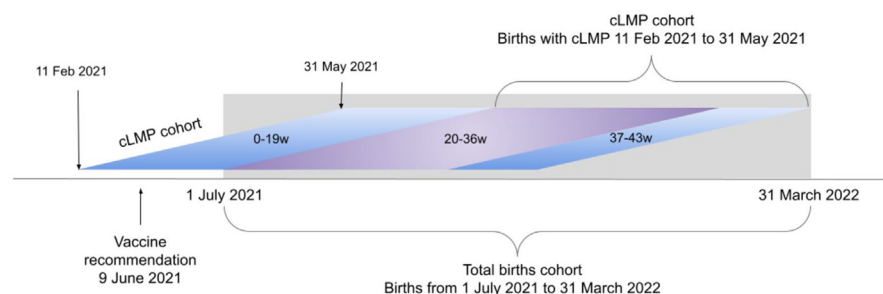
Outcome measures

Our analysis was performed with several denominator groups according to the outcomes of interest.

Total births cohort

Sociodemographic characteristics. We used the total number of women giving birth from July 1, 2021 to March 31,

FIGURE 1
Total births and calculated last menstrual period cohorts



cLMP, calculated last menstrual period.

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2022 to compare the characteristics of those with and without at least 1 dose of a COVID-19 vaccine before or during pregnancy. The characteristics included: maternal age, body mass index (BMI) in kg/m², smoking status, need for an English interpreter, maternal region of birth, socioeconomic status (assigned by residential postcode), parity, diabetes mellitus (none, gestational, preexisting, not tested), influenza vaccination status, pertussis vaccination status, plurality, gestation at first antenatal visit ≤ 12 weeks, and geographic remoteness.

Vaccination status by the week of birth. To examine temporal patterns in the uptake of COVID-19 vaccine following the health authority recommendations, we measured the weekly proportion of births to women who had received ≥ 1 doses of the mRNA COVID-19 vaccine before or during pregnancy.

Choropleth maps of Socio-Economic Indexes for Areas (SEIFA) and vaccination rates were generated in Tableau (version 2022.3; Salesforce, San Francisco, CA). Data on the specific brand of vaccine were not collected, but the Pfizer COVID-19 vaccine was the most available during the study period.

Calculated last menstrual period cohort

We used the calculated calendar week of the last menstrual period (LMP), rather than calendar week of birth, to define the vaccine-exposed and unexposed groups to ensure that any woman birthing from 20 to 43 weeks' gestation would be

captured in the birth data, thus avoiding "fixed cohort bias."¹⁴ We subtracted the infant's gestational age at birth in completed weeks from the calendar week of birth to obtain the week of calculated LMP (cLMP). For example, a woman giving birth at 40 weeks' gestation in the calendar week starting at October 4, 2021 would have a cLMP 280 days earlier on December 28, 2020. Week 20 of gestation would start on May 17, 2021. This person would be excluded from the cohort because she would already be at 26 weeks' gestation at the start of the birth data collection period on of July 1, 2021.

Using this cLMP, we defined a cohort containing women for whom weeks 20 to 43 of gestation would have occurred after mandatory vaccination status reporting commenced on July 1, 2021. This included women whose cLMP occurred during the 16 weeks from February 11, 2021 to May 31, 2021 inclusive (Figure 1).

The pregnancies for which weeks 20 to 43 of gestation would have occurred during the birth data collection period (July 1, 2021–March 31, 2022) were selected for the primary and secondary outcome analysis to avoid fixed-cohort bias. Multifetal gestations and women who had preconception COVID-19 vaccination were excluded only from the singleton cLMP cohort.

Congenital anomalies at ≥ 20 weeks. The rate of major congenital anomalies was calculated among singleton infants at ≥ 20 weeks' gestation in women who had received at least 1 COVID-19

vaccination dose during pregnancy, and terminated pregnancies were also included.

Perinatal outcomes at ≥ 24 weeks. Perinatal outcomes were calculated from the singleton cLMP cohort following exclusion of congenital anomalies, terminated pregnancies, and births < 24 weeks' gestation. Calculation was performed using an "all births" denominator (live births and stillbirths). These outcomes included:

1. Stillbirths: total, term, and preterm < 37 weeks.
2. Preterm birth < 37 weeks: total, spontaneous, and iatrogenic. An iatrogenic birth was defined as any birth without spontaneous onset of labor (ie, induced labor or no labor).
3. Fetal growth restriction (FGR): defined as birthweight ≤ 3 rd centile using Australian population sex-specific birthweight charts.¹⁶
4. Apgar score < 7 at 5 minutes.
5. Special-care nursery admission.
6. Neonatal intensive care unit (NICU) admission.
7. Mode of birth: induction of labor, prelabor cesarean delivery, cesarean delivery after labor onset, unassisted vaginal birth, instrumental vaginal birth (forceps/vacuum).
8. Born before arrival: refers to the rate of planned hospital births that occur before arrival, including unplanned births at home, in transit, or other locations.
9. Severe postpartum hemorrhage: estimated blood loss ≥ 1000 mL.
10. Iatrogenic birth for fetal compromise: total, ≥ 37 weeks, < 37 weeks. An iatrogenic birth was an induction of labor or cesarean delivery before labor onset. Indications for induction of labor and cesarean delivery were coded according to the Australian Institute of Health and Welfare's Metadata Online Registry (METEOR), which defines fetal compromise as "suspected or actual fetal compromise, and intrauterine growth restriction."¹⁷ Any documentation of suspected FGR, antepartum abnormal cardiotocography,

“fetal distress” (without labor), reduced fetal movements, oligohydramnios, abnormal umbilical artery Doppler studies, or placental insufficiency prompted inclusion in this classification.

All outcomes other than congenital anomalies and stillbirths were also calculated using live births as the denominator. Iatrogenic births for fetal compromise were only calculated using the live births denominator.

COVID-19 infections during pregnancy were collected and analyzed for the main outcomes of congenital anomalies, stillbirth, and preterm birth according to vaccination status.

Statistical analysis

No sample size calculation was performed because this was a cohort defined by the vaccination period. Analyses of secondary outcomes were considered exploratory, and no adjustments for multiple comparisons were made. Continuous variables in both determinants and safety analyses were presented with mean and standard deviation. Categorical variables were presented as counts and percentages. Statistical analyses were conducted using Stata, version 17 (StataCorp, College Station, TX), and 2-sided *P* values <.05 were considered statistically significant.

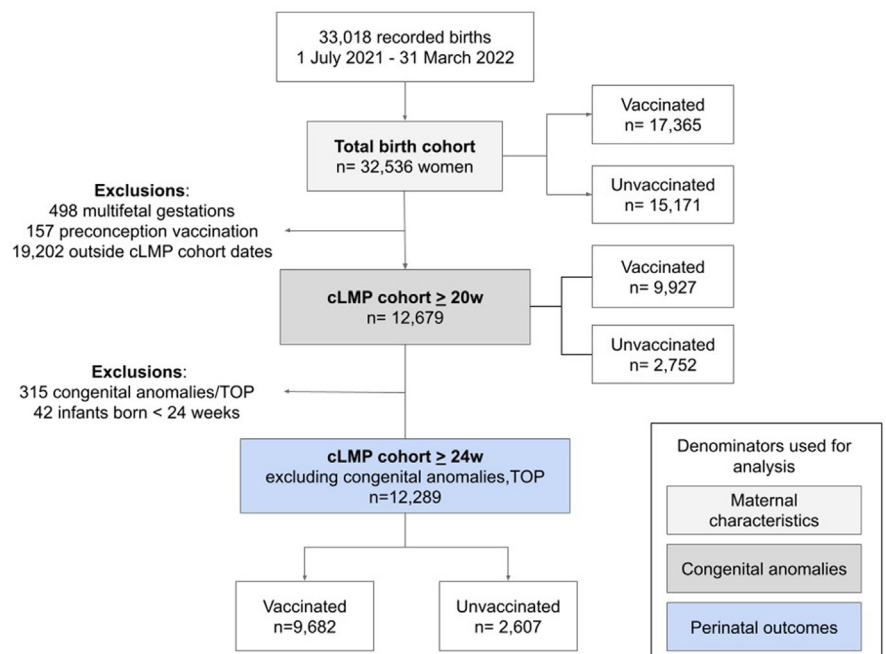
Determinants of COVID-19 vaccination uptake

We performed multiple imputation by chained equations to minimize the bias from missing data in our dataset and created 5 imputed datasets. Factors associated with antenatal COVID-19 vaccination were analyzed using multivariable Poisson regression, with covariates selected on the basis of subject matter knowledge. Results of the regression analyses were presented as incidence rate ratio. The hospital-specific rates of vaccination uptake were summarized and reported as forest plots.

Perinatal outcomes

We used inverse-probability-weighted regression adjustment (IPWRA) of the

FIGURE 2
Flowchart of study cohorts



cLMP, calculated last menstrual period; TOP, termination of pregnancy.

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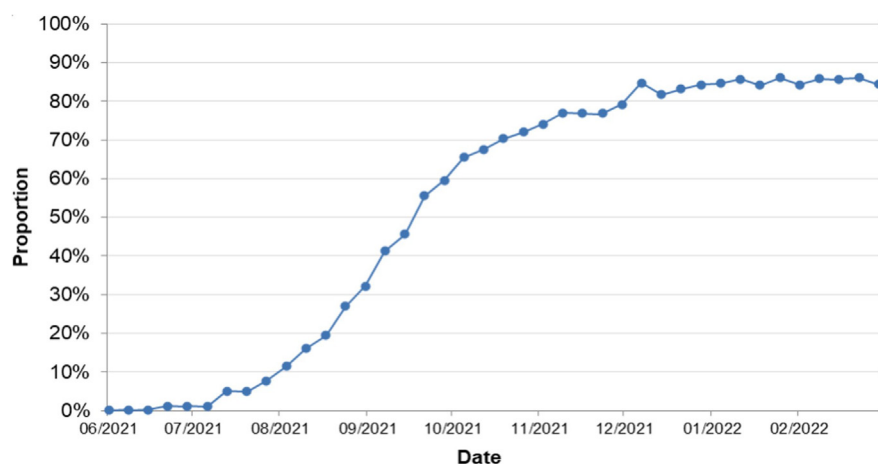
Stata “teffects” suite of commands to balance the baseline difference in the population by COVID-19 vaccination status. The IPWRA also accounts for the missing data. We adjusted for the following covariates: maternal age, BMI at first antenatal visit, maternal region of birth, need for interpreter (proxy indicator for primary language and categorized as “yes” or “no”), parity, socioeconomic status (using quintiles of SEIFA for maternal postcode), diabetes mellitus status (non, gestational, preexisting, not tested), and smoking in pregnancy status. These covariates were chosen for their known impact on the risk of adverse pregnancy outcomes (maternal age, BMI, parity, diabetes mellitus status, smoking, maternal region of birth) or their association with access to public health information and engagement in antenatal care (socioeconomic status, need for interpreter, maternal region of birth). Adjusted analyses were performed using logistic regression, and results were presented as odds ratios (ORs).

Sensitivity analyses

We performed unadjusted and adjusted logistic regression analysis without IPWRA to check the consistency of the directions of associations between COVID-19 vaccination and maternal and perinatal outcomes.

We performed a sensitivity analysis for congenital anomalies by excluding women who were first vaccinated at ≥ 20 weeks because the biologically relevant period of exposure for teratogenesis is in early pregnancy.

To examine the impact of time-varying exposure and the “healthy vaccinee” effect on preterm birth and stillbirth, we performed a Cox regression analysis excluding women who received their first dose of a COVID-19 vaccine at ≥ 24 weeks’ gestation, and adjusted for pertussis vaccination status. We used Kaplan–Meier curves to plot the cumulative hazard of the outcomes of interest. The proportionality of the hazards of control and exposed cohorts was tested using Schoenfeld residuals.

FIGURE 3**Weekly percentage of births to vaccinated women, July 2021–March 2022**

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Results

There were 33,018 infants born to 32,536 women during the study period. The numbers of inclusions and exclusions for the various cohorts are shown in the study flowchart in [Figure 2](#). The weekly percentages of births to vaccinated women are shown in [Figure 3](#). By the end of the study period, 85% of women giving birth had received at least 1 dose of the COVID-19 vaccine before or during pregnancy ([Figure 3](#)).

The weekly percentages of births to women with COVID-19 infection during pregnancy are shown in [Figure 4](#), representing a cumulative total of 1078

infected women. By the end of March 2022, 13% of women giving birth had experienced a COVID-19 infection during their pregnancy.

Sociodemographic characteristics of vaccinated vs unvaccinated groups

Of the 32,536 women giving birth during the 9-month study period, 17,365 (53.4%) had received at least 1 dose of the mRNA COVID-19 vaccine and 15,171 (46.6%) had not. The characteristics of vaccinated and unvaccinated groups are shown in [Table 1](#). Vaccinated women were significantly older, and more often

nulliparous, nonsmoking, not requiring an interpreter, of higher socioeconomic status, and vaccinated against pertussis and influenza ($P<.05$ for all variables). Vaccination status also varied by region of birth. Choropleth maps of postcodes showing socioeconomic status and vaccination coverage are shown in [Supplemental Figure 1](#). Forest plots of vaccination coverage by hospitals are provided in [Supplemental Figure 2](#).

Perinatal outcomes in the calculated last menstrual period cohort

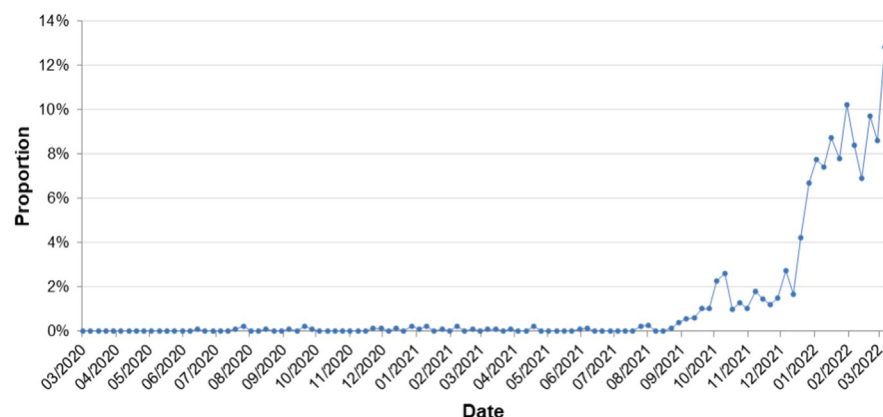
After exclusions, 12,679 participants remained in the cLMP cohort ([Figure 2](#)). Of these, 9927 had received at least 1 dose of the COVID-19 vaccine during pregnancy, and of these, 94.1% also received a second dose. A documented gestation at first and second vaccine dose was available for 98.3% and 92.7% of the vaccinated groups, respectively. The median gestational age at the first dose was 24 weeks (interquartile range, 19–28 weeks) and 28 weeks (interquartile range, 24–32 weeks) for the second dose ([Figure 5](#)). No data on third or subsequent vaccine doses were available.

The perinatal outcomes with “all births” and “live births” denominators are presented in [Tables 2](#) and [3](#), respectively.

Primary outcomes

The vaccinated group had a significantly lower rate of stillbirth compared with the unvaccinated group (0.2% vs 0.8%; adjusted OR [aOR], 0.18; 95% confidence interval [CI], 0.09–0.37; $P<.001$) ([Table 2](#)). When stratified by gestational age, this difference was statistically significant only for preterm stillbirths (0.1% vs 0.7%; aOR, 0.12; 95% CI, 0.05–0.31; $P<.001$). This finding remained robust in the sensitivity analysis ([Supplemental Table](#)).

The vaccinated group also had a significantly lower rate of preterm birth (5.1% vs 9.2%; aOR, 0.60; 95% CI, 0.51–0.71; $P<.001$), which was significant for both spontaneous preterm birth (2.4% vs 4.0%; aOR, 0.73; 95% CI, 0.56–0.96; $P=.02$) and iatrogenic preterm birth (2.7% vs 5.2%; aOR, 0.52;

FIGURE 4**Weekly rate of births to women with COVID-19 infection during pregnancy**

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TABLE 1
Maternal characteristics by COVID-19 vaccination status among total birth cohort

Maternal characteristics	Vaccinated N=17,365		Unvaccinated N=15,171		Multiple imputed dataset		
	n	%	n	%	Denominator	IRR ^a	Pvalue
Maternal age group, y							
<25	1477	8.5	1915	12.6	32,536	0.91	.005
25–29	3223	18.6	3377	22.3		Ref	
30–34	7224	41.6	5823	38.4		1.12	<.001
35–39	4487	25.8	3327	21.9		1.18	<.001
≥40	954	5.5	729	4.8		1.19	<.001
Parity							
0	7783	44.8	6259	41.3	32,535	1.09	<.001
≥1	9582	55.2	8911	58.7		Ref	
Plurality							
Singleton	17,102	98.5	14,936	98.5	32,536	Ref	
Multifetal pregnancy	263	1.5	235	1.6		1.00	.96
BMI categories							
<18	193	1.2	173	1.2	30,824	0.99	.90
18–24	7651	46.3	6477	45.3		Ref	
25–29	4892	29.6	4322	30.0		1.00	.97
30–34	2228	13.5	1950	13.6		1.03	.27
35–39	947	5.7	860	6.0		1.04	.26
≥40	603	3.7	528	3.7		1.08	.092
Smoking in pregnancy							
Nonsmokers	16,883	97.2	14,270	94.1	32,536		
Smokers	482	2.8	901	5.9		0.71	<.001
Region of birth							
Americas	317	1.8	209	1.4	32,352	1.05	.37
Australia	8916	51.7	7898	52.3		Ref	
North Africa and Middle East	661	3.8	829	5.5		0.89	.005
Northeast Asia	600	3.5	447	3.0		1.01	.76
Northwest Europe	616	3.6	373	2.5		1.09	.050
Oceania	527	1.6	633	4.2		0.90	.018
Southeast Asia	1525	8.8	1042	6.9		1.12	<.001
Southern and Central Asia	3426	19.9	2758	18.3		1.03	.17
Southern and Eastern Europe	249	1.4	390	2.6		0.71	<.001
Sub-Saharan Africa	426	2.5	510	3.4		0.89	.022
Interpreter required							
Not required	16,645	95.9	14,413	95.0	32,536		
Required	720	4.2	758	5.0		0.96	.27
SEIFA quintile							

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(continued)

TABLE 1

Maternal characteristics by COVID-19 vaccination status among total birth cohort (continued)

Maternal characteristics	Vaccinated N=17,365		Unvaccinated N=15,171		Multiple imputed dataset		
	n	%	n	%	Denominator	IRR ^a	Pvalue
1 (most disadvantaged)	3288	19.1	3551	23.7	32,181	0.94	.005
2	2466	14.4	2223	14.8		1.00	.96
3	4283	24.9	3899	26.0		Ref	
4	3980	23.2	3124	20.8		1.05	.039
5 (most advantaged)	3159	18.4	2208	14.7		1.07	.007
Remoteness of residence							
Regional	304	1.8	358	2.4	32,536	Ref	
Metropolitan location	17,061	98.3	14,813	97.6		0.93	.19
Gestation at first antenatal visit							
≤12 wk	12,252	70.6	10,656	70.2	32,536	Ref	
>12 wk	5113	29.4	4515	29.8		0.99	.58
Antenatal influenza vaccine							
Nonvaccinated	3815	22.2	4899	32.9	32,081	Ref	
Vaccinated	13,362	77.8	10,005	67.1		1.24	<.001
Antenatal pertussis vaccine							
Nonvaccinated	1415	8.2	3114	20.8	32,280	Ref	
Vaccinated	15,860	91.8	11,891	79.3		1.74	<.001
Diabetes mellitus							
None	13,012	75.0	11,555	76.2	32,527	Ref	
Gestational diabetes mellitus	4088	23.6	3299	21.8		1.03	.10
Preexisting diabetes mellitus	197	1.1	143	0.9		1.10	.17
Not tested for gestational diabetes mellitus	64	0.4	169	1.1		0.61	<.001

BMI, body mass index; IRR, incidence rate ratio; SEIFA, Socio-Economic Indexes for Areas.

^a IRR adjusted for maternal age, maternal smoking status, need for interpreter, socioeconomic index for areas, country of birth, parity and diabetes mellitus status.

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95% CI, 0.41–0.65; $P<.001$). These findings remained robust in the sensitivity analysis (Supplemental Table). The hazard ratio plots for preterm birth and stillbirth that excluded women vaccinated at ≥ 24 weeks, adjusted for pertussis vaccination status, are shown in Figure 6.

Secondary outcomes

The rate of major congenital anomalies was significantly lower in the vaccinated group than in the unvaccinated group (2.4% vs 3.0%; aOR, 0.72; 95% CI, 0.56–0.94; $P=.02$) (Table 2). This did not remain statistically significant in the sensitivity analysis that excluded women

vaccinated at ≥ 20 weeks' gestation, although the direction of the trend was similar (2.6% vs 3.0%; aOR, 0.80; 95% CI, 0.57–1.13; $P=.21$) (Supplemental Table).

There was also a significant reduction in admissions to the NICU in the vaccinated group (2.4% vs 4.1%; aOR, 0.70; 95% CI, 0.53–0.91; $P=.007$). This trend toward lower NICU admissions was observed in the sensitivity analysis, but with a P value $>.05$.

There was no difference in the rate of severe FGR between the vaccinated and unvaccinated groups. The vaccinated group had a significantly higher rate of induction of labor and iatrogenic births

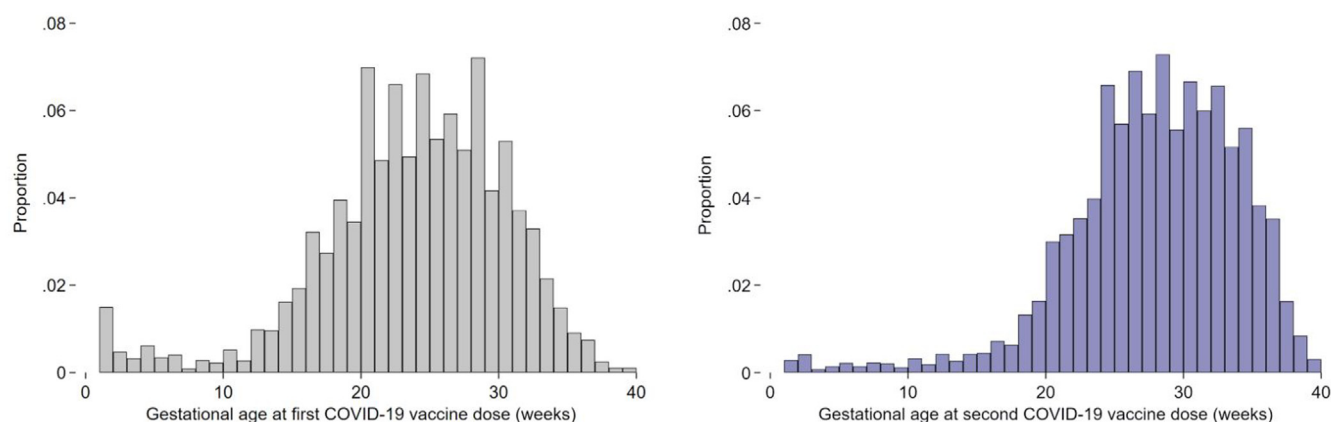
for fetal compromise at term gestation, but significantly lower rate of iatrogenic birth for fetal compromise at preterm gestations and lower rate of prelabor cesarean deliveries (Table 3). There was no significant difference in other birth outcomes such as postpartum hemorrhage, vaginal births, or births before arrival to hospital.

Subgroup analysis of stillbirth and preterm birth by COVID-19 infection during pregnancy and vaccination status

There were 1078 women with a COVID-19 infection during pregnancy in the total study cohort; of these, 518 women

FIGURE 5

Timing of first and second COVID-19 vaccine doses during pregnancy



Calculated on the singleton calculated last menstrual period cohort >20 weeks' gestation.

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were in the cLMP cohort (Table 4). The crude OR of COVID-19 infection was significantly lower in the vaccinated group (3.6% vs 5.9%; OR, 0.59; 95% CI, 0.49–0.71; $P<.0001$). There was also a significant reduction in preterm birth among COVID-19–infected women who had been vaccinated during pregnancy (2.3% vs 7.0%; OR, 0.32; 95% CI, 0.12–0.80; $P=.015$) (Table 4). The number of infected cases was too low to allow adjustment for covariates.

Comment

Principal findings

Our multicenter cohort study confirms the safety and benefits of COVID-19 vaccination in pregnancy and adds new insights into the relationship between vaccination and preterm birth. Vaccinated women had significantly lower rates of stillbirth and preterm birth compared with unvaccinated women and did not have higher rates of congenital anomalies or FGR. Of note, our study shows that COVID-19 vaccination during pregnancy is associated with significantly lower rates of both spontaneous and iatrogenic preterm birth, even after excluding women vaccinated at ≥ 24 weeks' gestation.

Results in the context of what is known

Our finding on congenital anomalies provides additional reassuring safety

data on COVID-19 vaccination in pregnancy. We have strengthened the existing evidence base by showing that there was no increase in the rate of congenital anomalies for women vaccinated before 20 weeks' gestation, when the risk of teratogenesis would be expected to be higher.

The significantly lower rate of stillbirths is in accordance with previously published studies, but our results are remarkable for the magnitude of the difference. Our aOR of 0.18 is much lower than that of any of the 7 individual studies included in the recent meta-analysis, which reported ORs of 0.50 to 1.50.¹² This difference was even more profound in the preterm stillbirths, in which vaccination was associated with a 1-in-1000 risk of preterm stillbirth as opposed to 7-in-1000 without vaccination.

It makes biological sense that vaccination reduces stillbirth given the strong association between severe COVID-19 infection and increased risk of stillbirth.¹ However, because of the observational study design, we can only infer a causal relationship between vaccination and fewer stillbirths, presumably mediated by the significant reduction in the odds of COVID-19 infection. However, the burden of COVID-19 infection in the unvaccinated cohort did not seem high enough to account for the entire effect of vaccination on stillbirths. Although we

controlled for multiple covariates, and performed a sensitivity analysis on pertussis vaccination status, we speculate that other relevant unmeasured factors associated with COVID-19 vaccination persist, such as health literacy or health care-seeking behavior (the “healthy vaccinee” bias).¹⁸ The significantly higher rates of term induction of labor and better adherence to universal gestational diabetes mellitus screening in the vaccinated group suggest important variations in obstetrical care that may have contributed to the difference in perinatal outcomes.

Our study provides strong evidence to support the association between COVID-19 vaccination and reduction in iatrogenic and spontaneous preterm births. The reduction in iatrogenic preterm birth may be because of prevention of complications directly related to COVID-19 infection such as preeclampsia or severe maternal respiratory morbidity, whereas the reduction in spontaneous preterm birth may be related to avoiding the systemic inflammation associated with COVID-19 infection or other nonspecific immune effects. Furthermore, unmeasured confounders may also be responsible for the reduction in preterm births, as discussed above for stillbirths.

The sociodemographic characteristics of our unvaccinated cohort mirror the findings from the United Kingdom

TABLE 2

Primary and secondary outcomes among all births (live births and stillbirths)

Primary and secondary outcomes	Vaccinated		Unvaccinated		Unadjusted odds ratio (IPWRA)				Adjusted odds ratio ^a (IPWRA)			
	n	(%)	n	(%)	OR	L	U	Pvalue	aOR	L	U	Pvalue
Singleton cLMP cohort ≥20 wk	N=9927		N=2752									
Congenital anomalies	236	2.4	83	3.0	0.79	0.62	1.01	.06	0.72	0.56	0.94	.02
Singleton cLMP cohort excluding congenital anomalies, TOP, and births <24 wk	N=9682		N=2607									
Stillbirths												
Stillbirths	15	0.2	21	0.8	0.19	0.10	0.37	.00	0.18	0.09	0.37	<.001
Stillbirths >37 wk	6	0.1	3	0.1	0.54	0.13	2.15	.38	0.38	0.09	1.55	.18
Stillbirths <37 wk	9	0.1	18	0.7	0.13	0.06	0.30	.00	0.12	0.05	0.31	<.001
Preterm birth <37 wk												
Total	495	5.1	239	9.2	0.56	0.48	0.65	.00	0.60	0.51	0.71	<.001
Spontaneous	234	2.4	104	4.0	0.61	0.48	0.76	.00	0.73	0.56	0.96	.02
Iatrogenic	261	2.7	135	5.2	0.52	0.42	0.64	.00	0.52	0.41	0.65	<.001
Newborn outcomes												
Fetal growth restriction	237	2.5	63	2.4	1.01	0.77	1.33	.93	1.07	0.79	1.44	.68
SCN admission	907	9.4	281	10.8	0.87	0.77	0.99	.03	0.92	0.80	1.06	.27
NICU admission	236	2.4	106	4.1	0.60	0.48	0.75	.00	0.70	0.53	0.91	.01
5-min Apgar score <7	170	1.8	80	3.1	0.57	0.44	0.74	.00	0.72	0.51	1.01	.06
Intrapartum outcomes												
Induction of labor	3589	37.1	835	32	1.16	1.09	1.23	.00	1.11	1.04	1.18	.001
Unassisted vaginal births	4738	48.9	1378	52.9	0.93	0.89	0.97	.00	0.98	0.94	1.02	.38
Instrumental vaginal births	1533	15.8	338	13.0	1.22	1.09	1.36	.00	1.08	0.96	1.21	.22
Cesarean delivery—no onset of labor	1829	18.9	529	20.3	0.93	0.85	1.02	.11	0.90	0.81	0.99	.04
Cesarean delivery —after onset of labor	1583	16.4	360	13.8	1.18	1.06	1.32	.00	1.07	1.07	1.21	.27
Born before arrival	62	0.6	27	1.0	0.62	0.39	0.97	.04	0.81	0.50	1.31	.39
Severe PPH ≥1000 mL	806	8.3	205	7.9	1.06	0.91	1.23	.45	0.97	0.83	1.13	.72

aOR, adjusted odds ratio; cLMP, calculated last menstrual period; IPWRA, inverse-probability-weighted regression adjustment; L, lower limit of 95% confidence interval; NICU, neonatal intensive care unit; OR, odds ratio; PPH, postpartum hemorrhage; SCN, special care nursery; TOP, termination of pregnancy; U, upper limit of 95% confidence interval.

^a Adjusted for maternal age, metropolitan vs regional residence, smoking status, need for interpreter, body mass index, region of birth, socioeconomic index for postcodes, diabetes mellitus, parity, infant sex, and gestation at first antenatal visit.

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showing variation in vaccination uptake by socioeconomic status, age, and region of birth.¹⁹ In multicultural Melbourne, public health messaging to non-English speaking communities had notable flaws, including incorrect and out-of-date translations of vaccine communications.²⁰ This may have contributed to a lack of access to accurate vaccine safety

information and a lack of trust in the health authorities among certain ethnic groups.

Clinical implications

Our data give us strong evidence to confidently endorse the safety of the mRNA vaccine for our local pregnant population and to promote the benefits

of reduced stillbirth and preterm birth. Our vaccine coverage of 85% at the end of March 2022 is an achievement that surpasses those recorded in the United States and the United Kingdom (70% and 60% in 2022, respectively).^{19,21} However, the substantial regional variation in vaccine coverage within Melbourne (Supplemental Figure 1) reflects

TABLE 3
Perinatal outcomes among live births

Singleton cLMP cohort excluding congenital anomalies, TOP, births <24 wk, and stillbirths	Vaccinated N=9667		Unvaccinated N=2586		Unadjusted odds ratio (IPWRA)				Adjusted odds ratio ^a (IPWRA)			
	n	(%)	n	(%)	OR	L	U	Pvalue	aOR	L	U	Pvalue
Preterm birth <37 wk												
Total	486	5.0	221	8.6	0.59	0.50	0.69	<.001	0.64	0.54	0.76	<.001
Spontaneous	230	2.4	102	3.9	0.60	0.48	0.76	<.001	0.72	0.55	0.94	.017
Iatrogenic	256	2.7	119	4.6	0.58	0.47	0.71	<.001	0.58	0.46	0.74	<.001
Newborn outcomes												
Fetal growth restriction	236	2.4	62	2.4	1.02	0.77	1.34	.899	1.08	0.79	1.46	.63
SCN admission	907	9.4	281	10.9	0.86	0.76	0.98	.023	1.08	0.80	1.06	.27
NICU admission	236	2.4	106	4.1	0.60	0.48	0.75	<.001	0.69	0.53	0.91	.01
5-min Apgar score <7	155	1.6	60	2.3	0.69	0.51	0.93	.013	1.39	0.51	1.01	.06
Intrapartum outcomes												
Induction of labor	3584	37.1	820	31.7	1.17	1.10	1.24	<.001	1.12	1.05	1.19	.001
Cesarean delivery—no onset of labor	1826	18.9	526	20.3	0.93	0.85	1.01	.094	0.90	0.81	0.99	.04
Unassisted vaginal births	4728	48.9	1360	52.6	0.93	0.89	0.97	.001	0.99	1.06	1.03	.50
Instrumental vaginal births	1531	15.8	338	13.1	1.21	1.09	1.35	.001	1.07	0.95	1.20	.27
Cesarean delivery—after onset of labor	1583	16.4	360	13.9	1.18	1.06	1.31	.003	0.30	0.94	1.20	.30
Born before arrival	61	0.6	26	1.0	0.63	0.40	0.99	.046	0.83	0.51	1.35	.45
Severe PPH ≥1000 mL	805	8.3	205	7.9	1.05	0.91	1.22	.512	0.96	0.83	1.13	.64
Iatrogenic births for fetal compromise												
Iatrogenic birth for fetal compromise	1853	19.17	451	17.44	0.46	0.34	0.64	.00	0.49	0.34	0.70	<.001
Iatrogenic birth for fetal compromise ≥37 wk	1751	18.11	392	15.16	1.19	1.08	1.32	.00	1.11	1.00	1.24	.040
Iatrogenic birth for fetal compromise <37 wk	102	1.06	59	2.28	0.46	0.34	0.64	.00	0.49	0.34	0.70	<.001

aOR, adjusted odds ratio; cLMP, calculated last menstrual period; IPWRA, inverse-probability-weighted regression adjustment; L, lower limit of 95% confidence interval; NICU, neonatal intensive care unit; OR, odds ratio; PPH, postpartum hemorrhage; SCN, special care nursery; TOP, termination of pregnancy; U, upper limit of 95% confidence interval.

^a Adjusted for maternal age, metropolitan vs regional residence, smoking status, need for interpreter, body mass index, region of birth, socioeconomic index for postcodes, diabetes mellitus, parity, infant sex, and gestation at first antenatal visit.

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the socioeconomic and cultural diversity of our population. A sensitive, multifaceted approach in partnership with community leaders is required to improve public health communication and remove barriers to vaccination uptake.²²

Research implications

The significant reduction in stillbirth and preterm birth raises research questions that may lead to improved care outside the pandemic. Biological and social factors responsible for the reduction in spontaneous preterm birth should be investigated, including

nonspecific inflammatory-mediated mechanisms for preterm birth and the mitigating effects of vaccination for COVID-19 and other diseases.

It is also likely that vaccination status is a proxy for unmeasured social determinants of health, in addition to the well-established demographic factors included in our analysis. This “healthy vaccinee bias” may reflect the quality of the relationship between a pregnant woman and her healthcare provider because personal recommendation by a healthcare practitioner remains one of the most important influences on vaccine uptake.¹¹ It is also correlated with

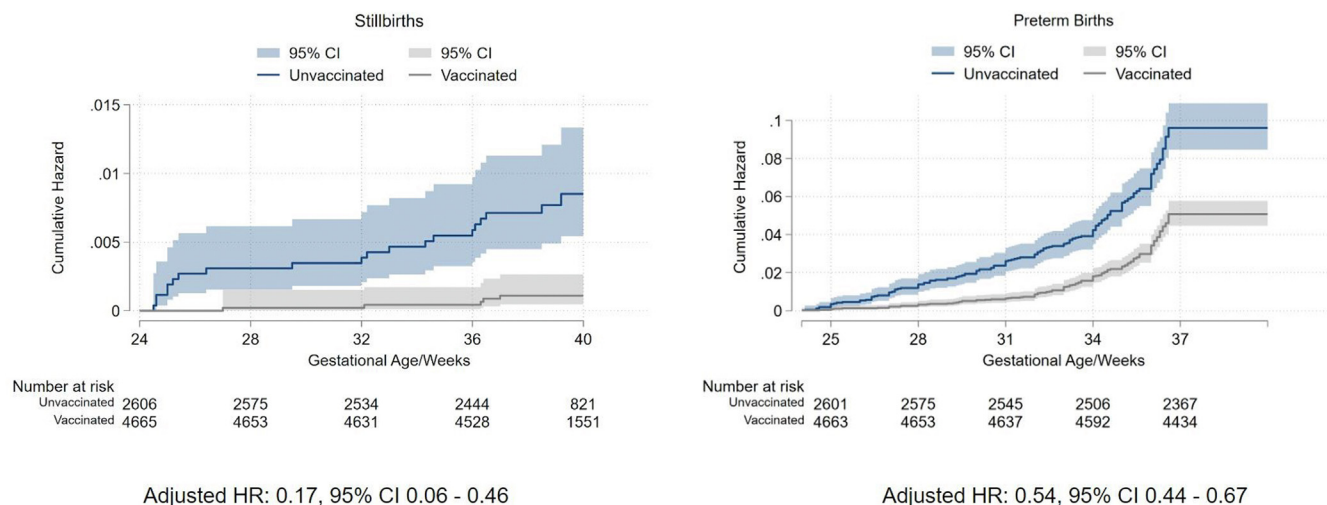
trust in the government.²³ Understanding these confounders and developing new metrics to capture them would aid future studies of vaccination in pregnancy and perinatal outcomes.

Strengths and limitations

Our large multicenter cohort of all public hospitals captured 80% of total births in Melbourne, including all hospitals that were designated by the government to care for pregnant inpatients with COVID-19. The major strengths of our cohort are its size, timeliness of data collection, avoidance of common methodological biases, and detailed

FIGURE 6

Hazard ratio plots for stillbirth and preterm birth for women vaccinated before 24 weeks' gestation



Plots were generated using the multiple imputed dataset and included adjustment for pertussis vaccination status.

CI, confidence interval; HR, hazard ratio.

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individual patient-level data including gestation at first vaccine dose.

This was a retrospective study using routinely collected maternity data. Only data on first and second doses of the COVID-19 vaccine were collected by hospitals during the study period, thus the uptake of the third dose by pregnant women is unknown. The 1078 women with COVID-19 infection in the total cohort should be considered a minimum estimate of

our total COVID-19 caseload. We do not have individual medical record data to determine whether the stillbirths and preterm births in the unvaccinated cohort were the direct result of acute COVID-19 disease. We await the outcomes from the CHOPAN (Coronavirus Health Outcomes in Pregnancy and Newborns) registry for detailed local information on pregnancy outcomes for women with COVID-19.²⁴

Our maternity data collection only includes birth outcomes from 20 weeks' gestation. We therefore cannot comment on risk of miscarriage after COVID-19 vaccination, although other large studies have already provided reassuring data on this.^{25,26} We acknowledge that our findings on the risk of congenital anomalies are limited by the lack of data on early pregnancy. The lower rate of congenital anomalies observed in the vaccinated group may be a result of

TABLE 4

Selected outcomes among women with documented COVID-19 infection by vaccination status

Outcomes	Vaccinated		Unvaccinated		Unadjusted odds			
	N=9927		N=2752		OR	L	U	Pvalue
	count	(%)	count	(%)				
Singleton cLMP cohort with COVID-19 infection	N=355		N=163					
Congenital anomalies	9	2.5	4	2.5	1.0	0.3	3.4	.96
Singleton cLMP cohort excluding congenital anomalies, TOP, and births <24 wk with COVID-19 infection	N=346		N=158					
Stillbirths	1	0.3	2	1.3	0.2	0.0	2.5	.23
Total preterm birth <37 wk	8	2.3	11	7.0	0.3	0.1	0.8	.02
Spontaneous preterm birth	1	0.3	4	2.5	0.1	0.0	1.0	.05
Iatrogenic preterm birth	7	2.0	7	4.4	0.5	0.2	1.3	.14

cLMP, calculated last menstrual period; L, lower limit of 95% confidence interval; OR, odds ratio; TOP, termination of pregnancy; U, upper limit of 95% confidence interval.

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Conclusions

Acknowledgments

- Mercy Health (Mercy Hospital for Women, Werribee Mercy Hospital)
- The Royal Women's Hospital, The Women's at Sandringham
- Monash Health (Monash Medical Centre, Casey Hospital, Dandenong Hospital)
- Northern Health (The Northern Hospital)
- Western Health (Joan Kirner Women's and Children's Hospital)
- Eastern Health (Box Hill Hospital, The Angliss Hospital)
- Peninsula Health (Frankston Hospital)

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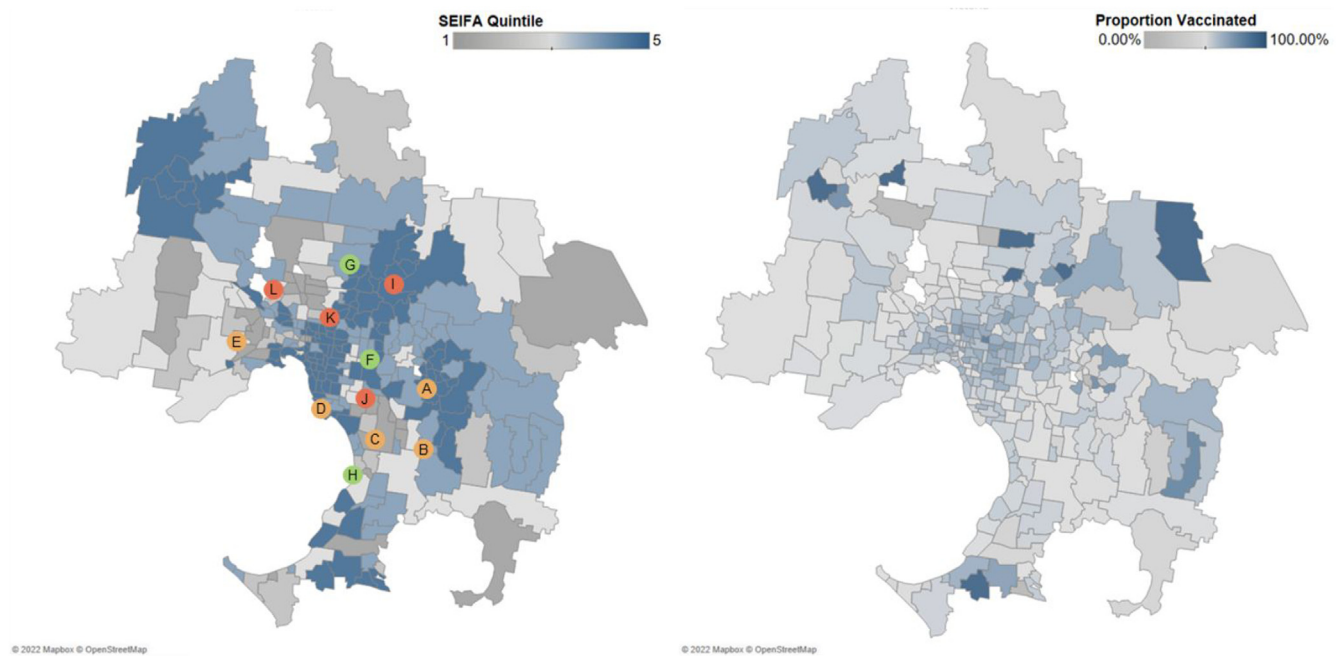
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The findings of this study were presented as a poster communication at the annual scientific meeting of the Royal Australian and New Zealand College of Obstetricians and Gynaecologists, Gold Coast, Australia, October 8–12, 2022.

Data sharing: Nonidentifiable individual participant data are available on request to the Austin Health Human Research Ethics Committee (ethics@austin.org.au) and the Mercy Health Human Research Ethics Committee (ethics@mercy.com.au). The study investigators may contribute aggregate and nonidentifiable individual patient data to national and international researchers whose proposed use of the data has been ethically reviewed and approved by an independent committee and following signing of an appropriate research collaboration agreement with The University of Melbourne.

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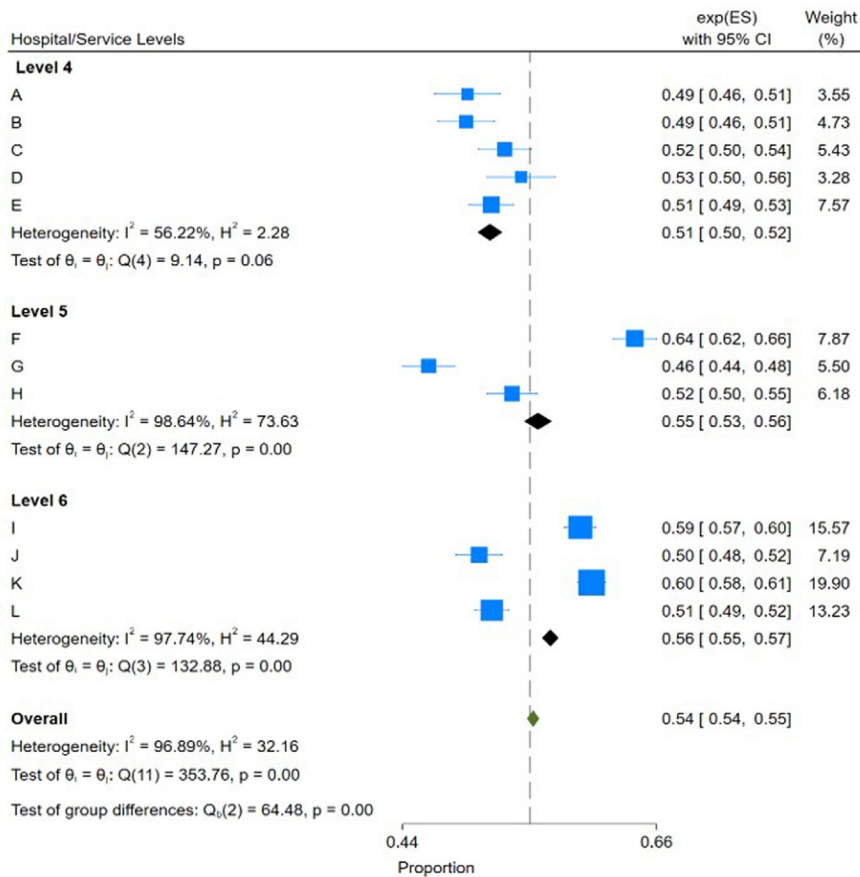
SUPPLEMENTAL FIGURE 1
Socio-Economic Indexes for Areas and maternal vaccination coverage by postcode



The participating hospitals are indicated by the lettered circles, with their corresponding vaccination rates displayed in [Supplemental Figure 2](#).
SEIFA, Socio-Economic Indexes for Areas.
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SUPPLEMENTAL FIGURE 2

Forest plots of vaccination coverage by hospital



The hospital-level data are grouped here by maternity service level. Level 6 maternity services provide regional/statewide specialized care for high-risk pregnancies, including extremely preterm births and local care for all women and infants; Level 5 services care for normal- to moderate-risk pregnancies, and manage labor and birth from 31 weeks' gestation; Level 4 services provide local care for women and infants at normal and moderate risk, including planned births from 34 weeks' gestation. The geographic location of each hospital is indicated in [Supplemental Figure S1](#).

CI, confidence interval.

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SUPPLEMENTAL TABLE

Sensitivity analysis for congenital anomalies, stillbirths, preterm births, and nursery admissions among all births

Outcomes	Vaccinated		Unvaccinated		Unadjusted odds ratio (IPWRA)				Adjusted odds ratio ^a (IPWRA)			
					OR	L	U	Pvalue	aOR	L	U	Pvalue
Singleton cLMP cohort vaccinated <20 wk	N=2442		N=2668									
	n	%	n	%								
Congenital anomalies	66	2.6	83	3.0	0.87	0.63	1.20	.400	0.80	0.57	1.13	.21
Singleton cLMP cohort vaccinated <24 wk excluding congenital anomalies, terminations of pregnancy, and births <24 wk	N=2442		n=4665									
	n	%	n	%								
Stillbirths												
Stillbirths	6	0.1	21	0.8	0.16	0.06	0.39	<.001	0.17	0.07	0.44	<.001
Stillbirths >37 wk	2	0.0	3	0.1	0.37	0.06	2.23	.28	0.22	0.04	1.39	.11
Stillbirths <37 wk	4	0.1	18	0.7	0.12	0.04	0.37	<.001	0.15	0.05	0.47	.001
Preterm birth <37 wk												
Total	231	5.0	239	9.2	0.54	0.45	0.64	<.001	0.58	0.48	0.71	<.001
Spontaneous	114	2.4	104	4.0	0.61	0.47	0.80	<.001	0.73	0.73	0.99	.046
Iatrogenic	117	2.5	135	5.2	0.73	0.54	0.99	.046	0.48	0.36	0.63	<.001
Newborn outcomes												
SCN admission	422	9.1	281	10.9	0.84	0.73	0.97	.016	0.90	0.77	1.05	.19
NICU admission	134	2.9	106	4.1	0.71	0.55	0.91	.006	0.81	0.61	1.09	.17

aOR, adjusted odds ratio; cLMP, calculated last menstrual period; IPWRA, inverse-probability-weighted regression adjustment; L, lower limit of 95% confidence interval; NICU, neonatal intensive care unit; OR, odds ratio; PPH, postpartum hemorrhage; SCN, special care nursery; TOP, termination of pregnancy; U, upper limit of 95% confidence interval.

^a Adjusted for maternal age, metropolitan vs regional residence, smoking status, need for interpreter, body mass index, region of birth, socioeconomic index for postcodes, diabetes mellitus, parity, infant sex, and gestation at first antenatal visit.

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